NOAH SEANER

Horsham, PA • 267-406-9360 • noahseaner@gmail.com • noah-seaner.github.io/ • linkedin.com/in/noah-seaner/

Summary

Data and sports analytics professional with experience in statistical modeling, business intelligence, database development, forecasting, optimization, and stakeholder reporting. Skilled in R, SQL, Excel, and Tableau, and quantitative decision-making; pursuing an M.S. in Business Intelligence & Analytics.

Education

Saint Joseph's University **(Expected July 2027)**
Master of Science in Business Intelligence & Analytics

Lafayette College **May 2025**
Bachelor of Arts in Mathematics and German (dual major): Concentration in Statistics, Minor: Data Science
GPA: 3.6

Technical Skills

- **Programming Languages:** R, SQL
- **Statistical Software & Libraries:** RStudio, tidyverse, nimble, coda, MCMCvis
- **BI & Visualization:** Tableau, Plotly, Excel dashboards
- **Statistical Modeling:** Regression Analysis, Bayesian Modeling, Hypothesis Testing
- **Forecasting & Simulation:** Moving Averages, Exponential Smoothing, Monte Carlo Simulation, Predictive Modeling
- **Optimization & Decision Analysis:** Linear Programming, Excel Solver, Goal Seek, Decision Trees
- **Database Management:** Microsoft Access, Relational Database Design, SQL Queries, Data Validation
- **Excel & Analytical Tools:** PivotTables, XLOOKUP, Data Analysis ToolPak, XLTree
- **Other Technologies:** FlightScope, GitHub, Google Sheets

Experience

All-Star Sports Academy **May 2026-present**
FlightScope Data Analyst Intern *West Chester, PA*

- Collect, validate, and analyze 5 key baseball performance metrics – exit velocity, launch angle, carry distance, pitch velocity, and spin rate – using FlightScope technology to support player evaluation and development.
- Operate and troubleshoot FlightScope tracking systems during 6 tournaments and 2 college showcases, ensuring consistent and accurate data collection throughout live competition.
- Capture and organize player-tracking data across an estimated 60-80 teams, supporting performance evaluation for hundreds of youth baseball players.
- Maintain accurate player-performance datasets that support the generation of performance summaries and allow coaches and athletes to evaluate development trends.
- Coordinate with coaches, event staff, and other personnel during full-day competitions to maintain data collection quality and resolve equipment or tracking issues.

Lafayette Men's Basketball **August 2024-May 2026**
Manager of Coaching Analytics *Easton, PA*

- Built and maintained multi-season player- and game-level datasets tracking 15+ Division I athletes over 30+ games per season using 20+ contextual game performance variables to support coaching decisions and statistical analysis.
- Developed regression models in R to evaluate player efficiency, scoring probabilities, and factors influencing team performance.
- Created Monte Carlo simulations to estimate individual-game win probabilities and conference standings, providing coaches with quantitative insights to identify challenging matchups and prioritize game preparation.
- Produced weekly interactive shooting dashboards in R/Plotly and prepared postgame analytical reports, delivering approximately 1-4 reports per week to support performance evaluation and coaching decisions.
- Automated Excel and R data collection workflows, reducing manual processing time by 50% and improving the timeliness of analytical reporting.
- Managed and trained 7 undergraduate student analysts, coordinating assignments, monitoring deadlines, and validating collected statistics to maintain data accuracy and consistency.

Global Career Accelerator **May 2024-July 2024**
Excel & Tableau Trainee *Horsham, PA (remote)*

- Completed 16 distinct business analytics projects using Excel and Tableau, analyzing datasets from different organizations to identify performance trends and develop actionable recommendations.
- Developed interactive dashboards to evaluate consumer behavior, website engagement, and operational performance.
- Applied A/B testing, statistical hypothesis testing, and exploratory data analysis to evaluate business performance.
- Cleaned, transformed, and aggregated datasets from multiple sources using Excel to improve data consistency and analytical readability.
- Presented findings through interactive visualizations, statistical summaries, and business recommendations to communicate insights to non-technical audiences.

Selected Research & Analytics Projects

Hillside Enterprises – Business Analytics & Decision Optimization
Excel • Forecasting • Monte Carlo Simulation • Decision Trees • Linear Programming

- Compared three forecasting models – moving averages, exponential smoothing, and linear regression – identifying a 3-month moving average with the lowest reported mean squared error (201,432.5) and forecasting approximately 6,681 January shipments.
- Conducted 100 simulations across three transportation routes, identifying the route with the shortest average travel time (24.41 minutes) and the fastest result in 48% of simulations.
- Applied decision-tree and expected-value analysis to compare technology investments, then used linear programming to optimize 72 production lines, achieving a modeled maximum revenue of $105,300 per hour.

Schmetsy Database – Relational Database Design & SQL Development
Microsoft Access • SQL • Relational Database Management

- Designed and developed a hypothetical relational database containing five interconnected tables and four relationships to organize customer, order, product, and shipping information.
- Wrote 60+ SQL queries to create and populate database tables, establish primary and foreign keys, update records, and retrieve information across related datasets.

Patriot League Basketball Season & Tournament Simulation
R • Monte Carlo Simulation • Probabilistic Forecasting

- Developed game-level simulations for a 10-team Division I conference, incorporating offensive and defensive efficiency, possession estimates, and game location to evaluate individual matchup win probabilities.
- Integrated game-level forecasts into season simulations to project conference records and standings, updating estimates throughout conference play; Lafayette ultimately finished one win and two standings positions above its initial projection.
- Ran 50,000 Monte Carlo tournament simulations to estimate championship probabilities, round advancement odds, and potential bracket outcomes.

Bayesian Analysis of MLB Offensive Performance
R • nimble • coda • MCMCvis • Bayesian Statistics

- Analyzed 18 MLB players across three offensive metrics – batting average, on-base percentage, and slugging percentage – to investigate changes in performance between 2024 and early 2025.
- Implemented Bayesian models using 20,000 MCMC iterations and a 5,000-iteration burn-in, estimating posterior distributions and credible intervals to quantify uncertainty in performance changes.
- Evaluated model convergence and summarized posterior estimates to assess evidence of changes in offensive production rather than relying exclusively on observed statistical difference.

Grammy Awards Website Analytics
Excel • Web Analytics • A/B Testing • KPI Analysis

- Analyzed approximately six years of website traffic and engagement data (2017-2023) across Grammy-related digital properties, examining visitors, sessions, page views, bounce rates, and session duration.
- Compared Grammy Awards-day traffic against typical daily traffic, evaluating audience behavior and engagement patterns associated with major entertainment events.
- Conducted statistical comparisons of website engagement metrics and benchmarked performance against a competing awards website to develop data-driven recommendations.

Intel Data Center Site Selection Dashboard
Tableau • Business Intelligence • Energy Analytics

- Analyzed 2022 energy production data across 13 regions and eight energy types to evaluate electricity availability, renewable energy utilization, and potential data center locations.
- Developed an interactive Tableau dashboard featuring three primary visualizations comparing regional net energy production, renewable energy percentages, and energy production trends over time.
- Incorporated filters for each region and energy type, as well as monthly reporting periods, enabling comparisons of regional energy profiles and sustainability considerations.

Publications/Presentations

- Featured in The Lafayette for developing large-scale simulation models used to project season and tournament outcomes.